1. Dental caries is a multifactorial process that includes the presence of 2 types of acid-producing Gram (+) bacteria present in dental plaque. A start up company is trying to develop a new treatment for patients with severe caries. Which of the below strategies should the company follow up on:
    a) A mouth wash containing small molecules that target Lipopolysaccharide
    b) A tooth paste containing molecules that promote bacterial adhesion to enamel
    c) A low pH gel that can be applied to the tooth surface to kill bacteria
    d) A rinse containing antibody-like molecules that bind to surface exposed adhesins on the Streptococci bacteria

2. What makes the Gram Stain a **differential** stain?
    a) Gram (-) cells have a LPS layer, which excludes Crystal Violet
    b) Gram (-) cells exclude Iodine (mordant) and get decolorized, where as Gram (+) cells allow entry of Iodine, which complexes with Crystal Violet
    c) Gram (+) cell walls exclude the ethanol-acetone decolorizer, whereas Gram (-) cells do not
    d) Gram (+) cells retain the crystal violet-iodine complex in the presence of decolorizer as compared to Gram (-) cells

3. You have isolated a gram-positive cocci from a skin lesion and want to determine if it is Staphylococcus aureus. Which one of the following laboratory test results would support a **definitive** identification as S. aureus?
    a) catalase positive
    b) coagulase positive
    c) gram (+) cocci in clusters
    d) beta-hemolysis on blood agar

4. With regard to bacterial genetics, “Transformation” describes a process that involves:
    a) The transfer of DNA through a cytoplasmic bridge
    b) The injection of viral DNA into bacterial cytoplasm
    c) The uptake of free DNA in the environment by competent bacterial cells
    d) The movement of of genetic material from one chromosomal locus to another

5. T helper cells augment B cell responses by:
    a) Recognizing antigen on the BCR
    b) Recognizing peptide displayed on an APC receptor
    c) Expressing Pattern Recognition Receptors to promote activation
    d) Expressing CD4 to provide co-stimulation

6. Your patient became ill 6 days ago with a viral disease. Laboratory examination reveals that the patient's antibodies against this virus have a high ratio of IgM to IgG. What is your conclusion?
   a) This patient encountered this organism recently
   b) This patient is predisposed to IgE-mediated hypersensitivity
   c) Your patient carries a mutation required for proper immunoglobulin class switching
   d) Your patient is not mounting a good response to the pathogen

7. You are studying a mouse with a non-functional TLR that detects flagellin. What outcome is expected if you inject purified flagellin into the mouse?
   a) Dendritic cells present flagellin on MHC II and activate helper T cells.
   b) Dendritic cells present flagellin on MHC I and activate cytotoxic T cells.
   c) B cells class switch to produce high-affinity IgG against flagellin.
   d) B cells recognize flagellin but do not receive T cell help.

8. The complement system plays a critical role in controlling bacterial infections. A major mechanism by which it achieves this is:
   a) Enzymatically digesting the cell membrane causing cell death
   b) Inducing macrophages to kill phagocytosed bacteria.
   c) Inserting pores into the cell membrane
   d) Inhibiting bacterial protein synthesis which leads to cell death

9. Which of the following best describes the process of antigen detection, transport, and presentation to T cells?
   a) Pathogens are phagocytosed by neutrophils, transported to lymph nodes, and presented to T cells via MHC class I.
   b) Antigens are endocytosed by dendritic cells, processed into peptides, and presented to T cells via MHC class II in the lymph nodes.
   c) B cells detect antigens in the bloodstream, transport them to the thymus, and present them to cytotoxic T cells via MHC class II.
   d) Macrophages detect pathogens in the lymph nodes, present antigens via MHC class I, and activate helper T cells.

10. Many bacteria produce a class of exotoxins, referred to as AB toxins, that act by a common mechanism. These toxins:
    a) are produced as subunits that when secreted, form pores in the cell membranes of their targets causing cell lysis
    b) are composed of lipid components that activate macrophages and induce a pro-inflammatory response
    c) interact with inflammatory mediators inside host cells causing apoptosis
    d) have target specificity and enzymatic activity that modifies host cell proteins, changing their function

11. In a T-cell dependent antibody response, which of the following T-cell properties is important for B-cell activation?
    a) IL-2 production
    b) gamma-interferon
    c) Expression of CD40 ligand
    d) Expression of B7

12. Standard precautions are a set of practices adopted to prevent the transmission of infectious diseases. Which of the following is a standard precaution used to prevent disease transmission?
    a) Using antiseptics to clean door handles and keyboards
    b) Disposing of all biohazard waste, including blood-contaminated materials, in designated containers.
    c) Performing hand hygiene after handling medical equipment in the client-care area.
    d) Properly removing PPE: After exiting the patient area, remove gloves, then goggles, then gown, and then use alcohol-based hand sanitizer.

13. Endotoxin is an important underlying cause of septic shock. The MOST accurate statement about endotoxin is that it:
    a) activates macrophages causing over expression of pro-inflammatory cytokines.
    b) is a polypeptide with an A-B subunit configuration.
    c) is found primarily in gram-positive organisms.
    d) can be chemically treated to form an effective toxoid vaccine.

14. Which one of the following is a function of antibody-mediated immunity?
    a) Antibodies help to activate the alternative pathway of complement
    b) Antibodies inhibit the growth of an intracellular bacterial pathogen
    c) Antibodies neutralize bacterial toxins
    d) Antibodies contribute to host tissue immunopathology of Delayed Type Hypersensitivity

15. Clonal selection describes the process by which:
    a) hematopoetic stem cells proliferate during development in the bone marrow
    b) an antigen induces B cell proliferation generating cells expressing antibody receptors with identical specificities
    c) B cells differentiate into an antibody secreting cells (plasma cell)
    d) gene segments randomly rearrange to generate diverse heavy and light chains

16. A 4 year-old child is stung by a bee and immediately experiences difficulty breathing, and lapses into unconsciousness. This reaction is MOST likely mediated by:
    a) IgA
    b) IgG
    c) IgE
    d) sensitized T- cells
    e) gamma-interferon

17. It is important to distinguish between nosocomial and community acquired Staphylococcal infections because:
    a) Community acquired Staph infections tend to have higher levels of drug resistance
    b) Community acquired Staph infections are more likely to cause serious respiratory infections
    c) Nosocomial Staph infections are more likely to be drug resistant
    d) Nosocomial Staph infections are more likely to present as serious soft tissue infections

18. Establishing immunological tolerance is essential for minimizing risk of developing autoimmune diseases caused by self-reactive T and B cells. Which of the below descriptions **BEST** describes a mechanism that leads to immune tolerance:
    a) Thymocytes with high affinity for self-MHC differentiate into regulatory T cells that are released into the periphery
    b) B cell somatic hyper-mutation is induced when an auto-reactive BCR is generated in the bone marrow.
    c) Self-reactive B cells in the periphery can undergo limited rounds of BCR rearrangement so they are no longer self-reactive
    d) Self-reactive B cells in the periphery that specifically interact with self antigen will not get T cell help and will undergo anergy

19. You are assisting in a dental exam on a patient who has an abscess on their gum. The dentist makes an incision and drains the abscess and then applies a substance to inhibit the infecting organism from re-growing. Which of these is the BEST choice to accomplish this:
    a) A high level disinfectant salve applied for 10 minutes
    b) An intermediate level disinfectant salve applied for 15 minutes
    c) A hydrogen peroxide-based antiseptic gel applied 3 times/day
    d) A topical broad spectrum antibiotic ointment applied as needed

20. Dendritic cells are critical to the development of an adaptive immune response. The major role they play is because they:
    a) are the primary cells that kill extracellular microbes
    b) recruit neutrophils and B cells to the site of infection
    c) present processed peptides and engage surface molecules to activate specific T cells
    d) process and secrete antigens to help activate B cells

21. How is antibody diversity generated?
    a) By variable, diversity and joining gene segment rearrangement
    b) By addition or loss of nucleotides at junctions between rearranging gene segments
    c) By random pairing of Ig heavy and light chains
    d) All of the above

22. Peripheral lymphoid organs:
    a) Include cutaneous and mucosal immune nodules where dendritic cells capture pathogens.
    b) Recognize pathogens that enter the bloodstream, gastrointestinal tract and lymph.
    c) Include MALT composed of Peyer's patches, tonsils and appendix.
    d) All of above.

23. The classical complement pathway is initiated by interaction of complement protein C1 with:
    a) antigen
    b) factor B
    c) antigen-IgG complexes
    d) bacterial lipopolysaccharide

24. What is a major effector mechanism of CD8 T cells?
    a) Releasing mediators that kill target cells
    b) Releasing cytokines that promote macrophage bactericidal activity
    c) Releasing chemokines to attract neutrophils to the site of infection
    d) Engaging TLR receptors on infected cells to induce interferon responses

25. Co-stimulation is crucial for activating naïve lymphocytes that results in acquisition of effector functions. Which of the below statements about co-stimulation is MOST accurate?
    a) Co-stimulation ensures the tight regulation of lymphocyte activation which helps to minimize the risk of autoimmunity
    b) Co-stimulation is required for the induction of peripheral tolerance
    c) Constitutive expression of co-stimulatory molecules facilitates swift activation when lymphocyte receptors encounter APCs expressing their cognate MHC:peptide
    d) Co-stimulation results in activation of lymphocytes and induction of anergy of the interacting APC

26. A young woman underwent rhinoplasty with postoperative nasal packing. Two days later, she developed fever, hypotension, a diffuse red rash, and chills. A nasal swab grew catalase (+), beta-hemolytic gram (+) cocci. What is the MOST likely bacterial culture?
    a) Streptococcus pyogenes
    b) Staphylococcus aureus
    c) Staphylococcus epidermidis
    d) None of the above

27. What is your diagnosis of the disease the patient in the previous question presented with?
    a) Scalded skin syndrome
    b) Toxic shock syndrome
    c) Carbunculosis
    d) Erysipelas
    e) Impetigo

28. Which of the following statements about IgG is the MOST accurate?
    a) Each IgG molecule has one antigen binding site.
    b) It has the most diversity of any immunoglobulin class.
    c) During the primary response, it is made in larger amounts than is IgM.
    d) It is the only immunoglobulin that is transferred from mother to fetus in utero.

29. A pediatric dental patient presents with recurrent oral abscesses. Genetic testing reveals a mutation preventing neutrophils from binding to endothelial integrins. Is this mutation relevant to the recurrent infections?
    a) No- the patient's tissue macrophages are functioning and should be able to control the infection
    b) No- dendritic cells at the site of infection will be activated through the TOL receptors and can activate T cells in the lymph nodes
    c) Yes- the mutation prevents neutrophils from becoming activated, so they are not sufficiently cytotoxin
    d) Yes- the mutation prevents neutrophils from accessing the infected tissue

30. T cells can respond to metals that induce delayed (type IV) hypersensitivity because the metals
    a) bind to costimulatory receptors.
    b) are processed and presented on MHC
    c) Induce antigen presenting cells to produce cytokines
    d) bind to self peptides, which are presented on MHC molecules
    e) bind directly to the TCR bypassing costimulation

31. Which of the following is most likely considered a mechanism of immune evasion by intracellular pathogens?
    a) Inhibition of phagolysosomal fusion
    b) Production of capsule
    c) Inhibition of complement
    d) Secretion of IgA proteases

32. Which of the following best describes the pathogenesis of rheumatic fever?
    a) An exotoxin produced by Streptococcus pyogenes acting as a superantigen damages the cardiac muscle.
    b) An exotoxin produced by Streptococcus pyogenes ADP-ribosylates a G protein causing damage to the cardiac muscle
    c) Antibody to a surface-localized protein of Streptococcus pyogenes cross-reacts with cardiac muscle resulting in complement activation
    d) Endotoxin produced by Streptococus pyogenes activates macrophages to release cytokines that damage the cardiac muscle

33. You are assisting in the clinic and are helping to clean up after a routine dental cleaning. After washing the instruments, which method of disinfection should be used for scalers?
    a) Boiling for 15 minutes, followed by 2 hours in a hot air oven set at 400°F
    b) Autoclaving
    c) Soak in 70% ethanol solution for 20 minutes
    d) Expose to a high level disinfectant for 10 minutes

34. A 5-year-old boy presents with recurrent Staphylococcus aureus boils. His pediatrician suspects an underlying immunological defect. Which of the following is MOST likely to explain his susceptibility?
    a) He produces defective C8 component of complement
    b) His macrophages are defective for antigen presentation on MHC I receptors
    c) His NK cells are defective for granzyme production
    d) He produces an insufficient amount of IgG

35. Which of the following occurs in the thymus:
    a) Processing and presentation of antigen from a peripheral infection on MHC II
    b) Rearrangement of gene segments that code for the variable region on antibodies
    c) T cell maturation
    d) B cell differentiation

36. A 6 year old girl is brought to the emergency room after stepping on a rusty nail. Her foot is sore, and she has no other symptoms. The caregiver who brought her is not sure whether the girl has received her tetanus vaccine. Based on what you have learned, which of the following is the **BEST** recommendation:
    a) Immunize her with the tetanus toxin
    b) Immunize her with the toxoid **and** administer immunoglobulins against tetanus
    c) Only administer immunoglobbulins against tetanus
    d) Since the girl does not have any symptoms that would suggest tetanus toxin exposure, treatment is not recommended

37. T cell development involves both positive selection and negative selection. Which of the following statements regarding these processes is correct:
    a) Negative selection is important for determining T cell receptor specificity
    b) Negative selection helps establish tolerance for self by deleting self-reactive T cells
    c) T cell receptors showing weak to no binding to MHC in the thymus will be positively selected
    d) Positive selection results in MHC class II expression on thymocyte surfaces whereas negative selection results in MHC class I expression on thymocyte surfaces

38. You isolate a pathogenic E. coli strain producing a toxin and expose it to UV light, causing cell lysis. The lysate (retaining nucleic acids) is incubated with non-pathogenic E. coli. After plating, the resulting colonies produce the toxin. Which horizontal transfer mechanism(s) explain this?
    a) Transformation or Conjugation
    b) Conjugation or Transduction
    c) Transformation or Transduction
    d) Conjugation or Meiosis

39. A patient is rushed to the hospital after suddenly developing fever, chills, hypotension (shock), and had signs of acute kidney failure. Blood cultures grow out a Gram (-) rod. The patient's symptoms are MOST likely due to the fact that the bacteria are:
    a) Invading the endothelium causing vascular leak
    b) Non-specifically engaging with CD4 T cells causing activation resulting in a cytokine storm
    c) producing endotoxin
    d) Producing a large amount of peptidoglycan that activates CD8 cells, causing systemic inflammation

40. A principal function of complement is to:
    a) Bind antibodies attached to microbial cell surfaces and to destroy these cells
    b) cross link allergens
    c) mediate the release of histamine
    d) phagocytize antigens

41. Streptococcus pyogenes, a Gram (+) extracellular cocci, can cause serious life threatening infections such as necrotizing fasciitis. Which of the below adaptive immune factors is MOST helpful for clearing this infection:
    a) CD8 T Cells
    b) Antibody-producing plasma cells
    c) MHC I presenting cells
    d) Natural Killer Cells

42. The expression of AIRE in thymic epithelial cells is important for:
    a) Determining the expression of CD4 or CD8 receptors on thymocytes
    b) The cortical epithelial cells to induce positive selection of TCRs
    c) Synthesizing tissue-specific proteins that can be then presented on MHC molecules
    d) Inducing self-reactive CD4 cells to become regulatory T cells

43. Hyper IgM syndrome is an X-linked immunodeficiency syndrome that leads to production of IgM, but little to no IgG or IgA. A mutation in which of the below molecules/pathways best explains the Hyper IgM syndrome phenotype :
    a) Mutation in CD40L. This results in T cells that cannot help B cells undergo class switching
    b) Mutation in B7. This results in B cells that cannot receive co-stimulation and thus don't produce IgG
    c) Mutation in Bruton's tyrosine kinase (BTK). This results in B cells with defective antigen receptor signaling
    d) Mutation in interferon signaling. This results in increased susceptibility to intracellular pathogens

44. Antibodies participate in hypersensitivity reactions in several ways. Which of the below statements is the MOST accurate with regard to antibody-mediated hypersensitivity reactions?
    a) IgE is involved in both anaphylactic and immune complex hypersensitivities
    b) Complement is involved in both anaphylactic and immune complex hypersensitivities
    c) Antigen:antibody complex deposition in tissues causes activation of complement and recruitment and degranulation of neutrophils.
    d) Neutrophils play a more important role in anaphylactic reactions than in immune complex reactions.

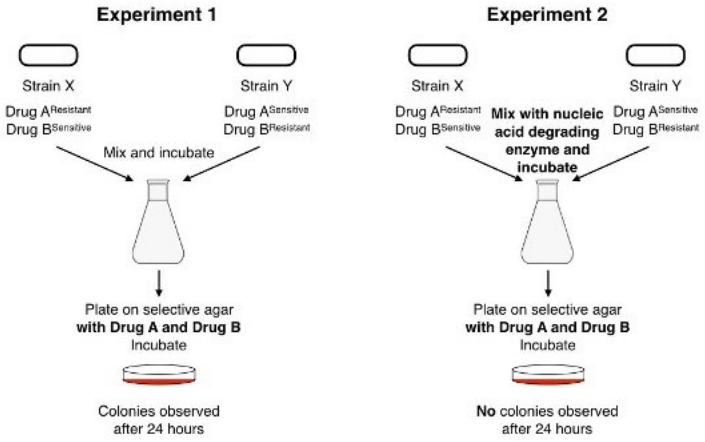
45. A 5 year old adopted child with X-linked agammaglobulinemia (XLA) has had several life-threatening infections. His medical team and parents decided that it was in the child's best interest to get a stem cell transplant. However, the closest HLA match was a non-biological sibling with only a 70% match. The major concern in proceeding with the transplant is:
    a) Many of the donor stem cells will develop into self-reactive T cells in the recipient, causing autoimmune disease
    b) The donor B cells will not be able to class switch to IgG
    c) The patient's new (donated) T cells won't be able to respond to peptide antigens presented by antigen presenting cells received from the donor
    d) The patient's antigen presenting cells will not be able to present peptides.

46. A 5 year old child, who has presented several times with sinusitis caused by different bacteria, now presents to clinic with bacterial pneumonia. Tests revealed that the child carried a mutation in the gene encoding CD40L. This mutation is associated with the recurrent infections because the child:
    a) is unable to activate naïve CD4+ T cells
    b) has fewer CD4+ T cells being positively selected in the thymus
    c) is unable to process and present antigens on dendritic cells
    d) is unable to make IgG or IgA

47. A year ago, your roommate had a bad cough and chest pain for a month. Now he has learned that he had an active Mycobacterium tuberculosis infection. He recommends you get a tuberculin skin test. At the doctor's office, they inject you intradermally with a protein extract of the bacteria. They tell you to come back in 2-3 days to be checked for the degree of inflammation at the site of injection. What component of the immune system is this test primarily dependent on?
    a) Complement
    b) Memory B cells
    c) Circulating antibodies
    d) Memory T cells
    e) Naive T cells

48. A new vaccine targets a virulent, encapsulated bacterium that infects the lungs and evades phagocytosis. The vaccine, made of polysaccharide capsule conjugated to a protein, induces a strong IgG response. Why is it effective?
    a) The capsule has a lipid component, which induces TLR signaling that allows dendritic cells to up regulate B7 for strong co-stimulation of T cells
    b) The strong antibody response induced by the vaccine allows for deposition of complement on infecting bacteria, which results in bacterial cell death
    c) The protein component of the vaccine allows dendritic cells to effectively activate CD4 cells, which promotes B cell differentiation
    d) The protein component of the vaccine provides signal 1 to T cell receptors that then receive co-stimulatory signals from dendritic cells

49. Strain X is resistant to Drug A but sensitive to Drug B. Strain Y is resistant to Drug B but sensitive to Drug A. After co-culture without antibiotics, colonies grow on agar with both drugs (Experiment 1). In Experiment 2, adding a nucleic acid-degrading enzyme during co-culture prevents colony growth. Assuming the enzyme cannot enter cells, which process explains these results?



a) Conjugation

b) Transduction

c) Transformation

d) Transposition

50. Babies born with Omenn syndrome have a mutation in RAG 1, which is essential for carrying out VDJ recombination in T and B cells. Based on this, which of the following is the MOST accurate statement:
    a) They will develop several T cell-related autoimmune diseases because their TCRs will not get negatively selected
    b) They will develop several B cell-related autoimmune diseases because receptor editing function is defective
    c) They will be able to make IgM, but cannot undergo isotope switching
    d) They can be cured by bone marrow transplant with an HLA matched donor

Patient
A 2-year-old patient with DiGeorge syndrome for a follow up visit after cleft palate surgery
Background and Patient History
The patient was diagnosed with DiGeorge syndrome at 8 months of age. He has an under-developed thymus (thymic hypoplasia), has hypocalcemia, and has been operated on for cleft palate. He has experienced 4 bouts of bacterial pneumonia in the last year.
Current Findings
Oral exam reveals that the child has several carious lesions and an abscess at the surgical site

51. You are a pediatric dental resident in a community clinic where this child is being seen. What information can you provide to the parents with regard to this child's condition:
    a) Patients with DiGeorge syndrome have T cell deficiency. A bone marrow transplant would cure this child's disease
    b) Based on the recurrent pneumonia, it is likely that the child has an underlying complement deficiency
    c) Children with DiGeorge syndrome have a defect in myeloid cell development and are unable to generate phagocytic cells
    d) Based on this child's history, he may benefit from IVIG therapy and antibiotic prophylaxis **CORRECT**